

Set 21: Energy Consumption Analysis (R Programming)

```
energy <- read.csv("Energy_Consumption_Data.csv")
```

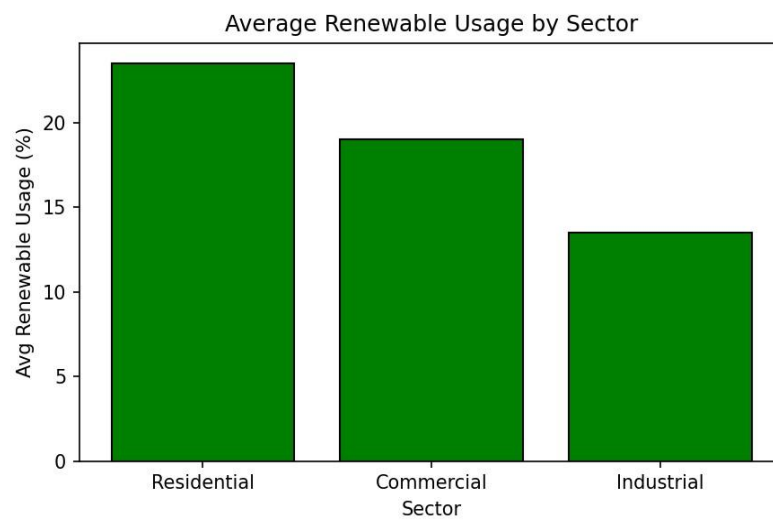
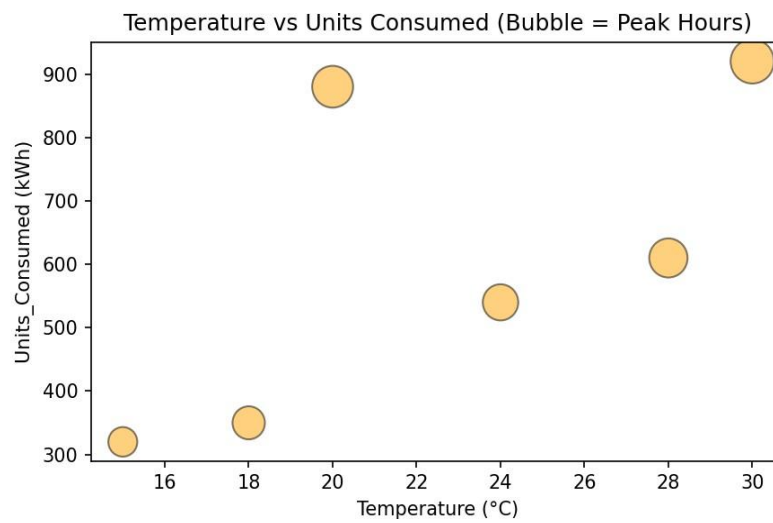
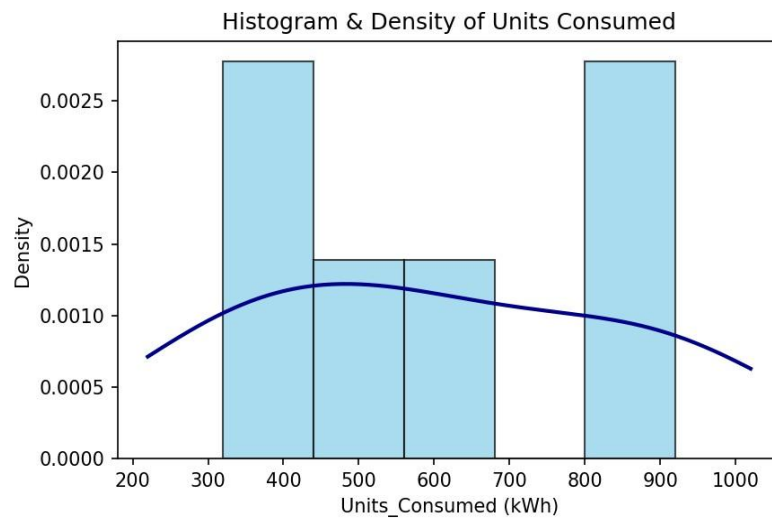
```
hist(  
  energy$Units_Consumed,  
  breaks = 5,  
  col = "skyblue",  
  probability = TRUE,  
  main = "Histogram & Density of Units Consumed",  
  xlab = "Units_Consumed (kWh)"  
)
```

```
lines(  
  density(energy$Units_Consumed),  
  col = "darkblue",  
  lwd = 2  
)
```

```
plot(  
  energy$Temperature,  
  energy$Units_Consumed,  
  cex = energy$Peak_Hours / 2,  
  pch = 19,  
  col = rgb(1, 0.5, 0, 0.5),  
  main = "Temperature vs Units Consumed (Bubble = Peak Hours)",  
  xlab = "Temperature (°C)",  
  ylab = "Units_Consumed (kWh)"  
)
```

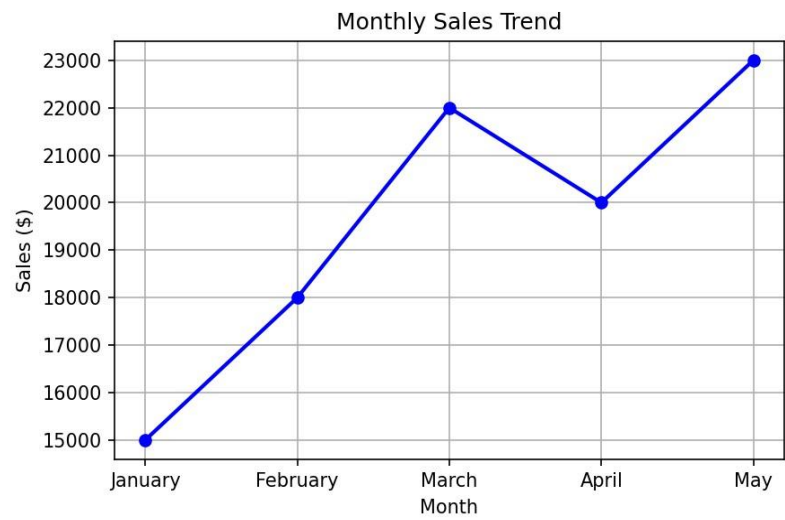
```
avg_renew <- aggregate(  
  Renewable_Usage ~ Sector,  
  data = energy,  
  FUN = mean  
)
```

```
barplot(  
  avg_renew$Renewable_Usage,  
  names.arg = avg_renew$Sector,  
  col = "forestgreen",  
  main = "Average Renewable Usage by Sector",  
  xlab = "Sector",  
  ylab = "Avg Renewable Usage (%)"  
)
```

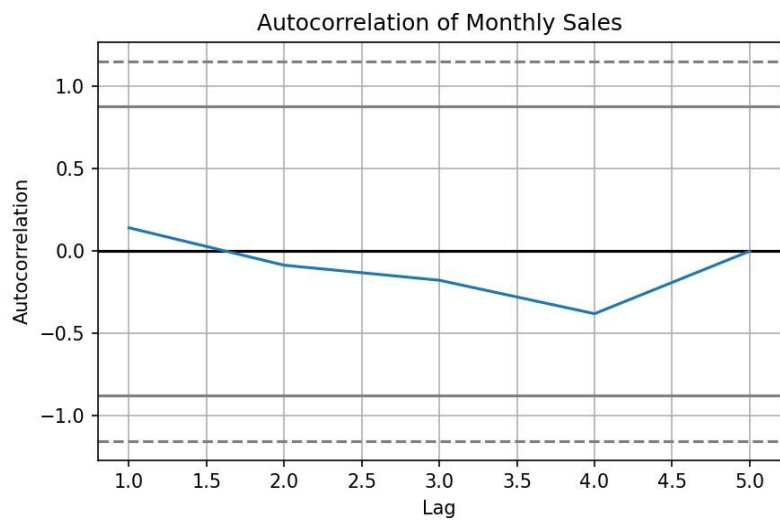
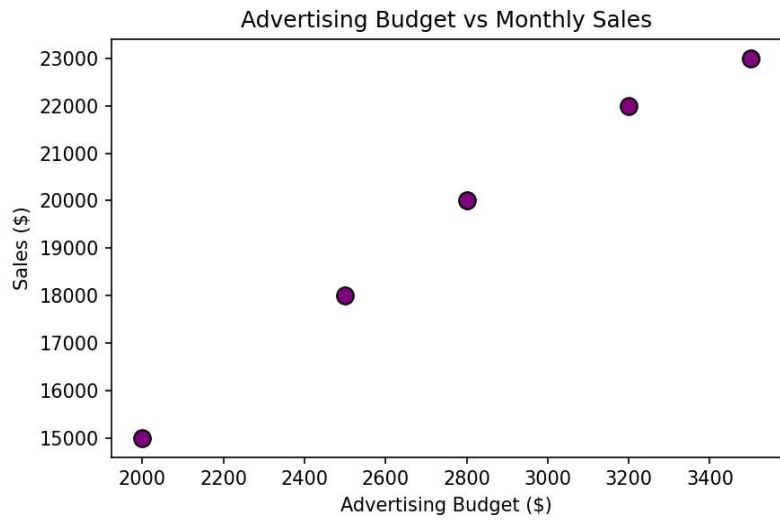


Set 22: Time Series Analysis (R Programming)

```
sales <- data.frame(  
  Month = c("January", "February", "March", "April", "May"),  
  Sales = c(15000, 18000, 22000, 20000, 23000)  
)  
  
sales$Month <- factor(sales$Month, levels = sales$Month)  
  
plot(  
  sales$Month,  
  sales$Sales,  
  type = "o",  
  col = "blue",  
  lwd = 2,  
  main = "Monthly Sales Trend",  
  xlab = "Month",  
  ylab = "Sales ($)"  
)  
  
grid()  
  
sales$Advertising_Budget <- c(2000, 2500, 3200, 2800, 3500)  
  
plot(  
  sales$Advertising_Budget,  
  sales$Sales,  
  pch = 19,  
  col = "purple",  
  main = "Advertising Budget vs Monthly Sales",  
  xlab = "Advertising Budget ($)",  
  ylab = "Sales ($)"  
)  
  
sales_ts <- ts(sales$Sales, frequency = 12)  
  
acf(  
  sales_ts,
```



main = "Autocorrelation of Monthly Sales"
)



Set 23: Employee Performance Analysis (R Programming)

```
emp <- data.frame(  
  Employee_ID = 1:5,  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  Years_of_Service = c(5, 3, 7, 4, 2),  
  Performance_Score = c(85, 92, 78, 90, 76)  
)
```

```
dept_count <- table(emp$Department)
```

```
barplot(  
  dept_count,  
  col = "teal",  
  main = "Employee Distribution by Department",  
  xlab = "Department",  
  ylab = "Number of Employees"  
)
```

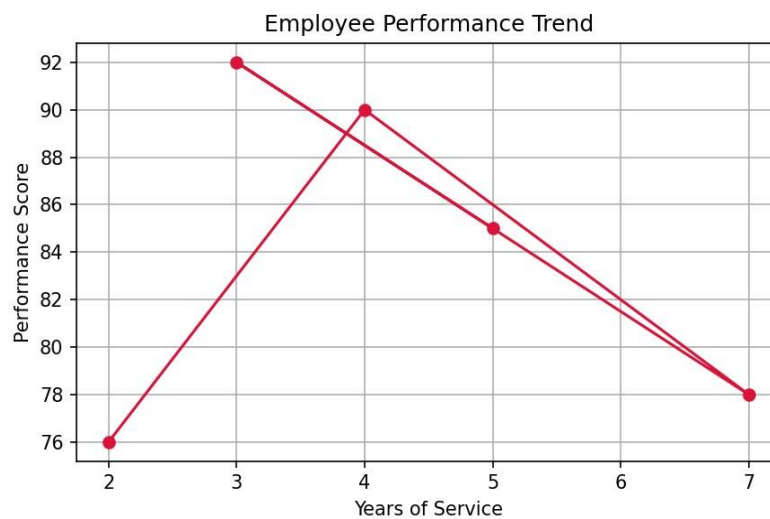
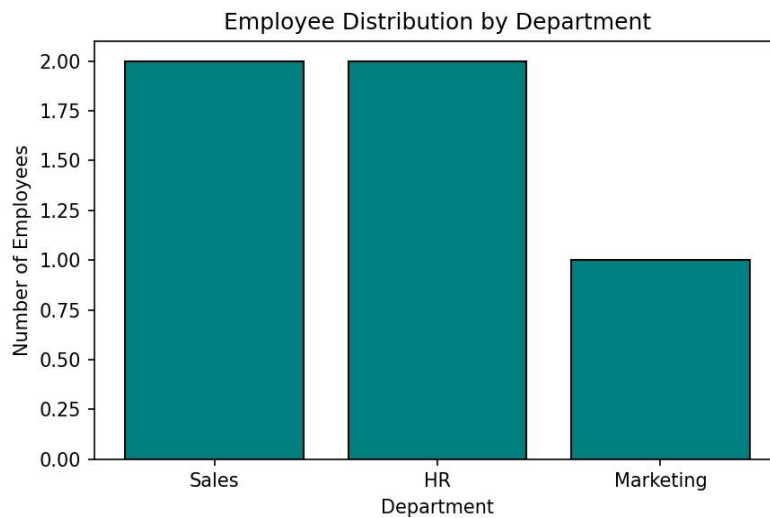
```
ord <- emp[order(emp$Years_of_Service), ]
```

```
plot(  
  ord$Years_of_Service,  
  ord$Performance_Score,  
  type = "o",  
  col = "crimson",  
  main = "Employee Performance Trend",  
  xlab = "Years of Service",  
  ylab = "Performance Score"  
)
```

```
grid()
```

```
library(gridExtra)
```

```
grid.table(emp)
```



Set 24: Product Inventory Management (R Programming)

```
inventory <- data.frame(
  Product_ID = 1:5,
  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity_Available = c(250, 175, 300, 200, 220),
  Category = c("Electronics", "Furniture", "Electronics", "Furniture", "Electronics"),
  Price = c(20, 15, 18, 22, 17)
)
```

```
barplot(
  inventory$Quantity_Available,
  names.arg = inventory$Product_Name,
  col = "skyblue",
  main = "Quantity of Products in Inventory",
)
```

```

xlab = "Product Name",
ylab = "Quantity Available"
)

```

```

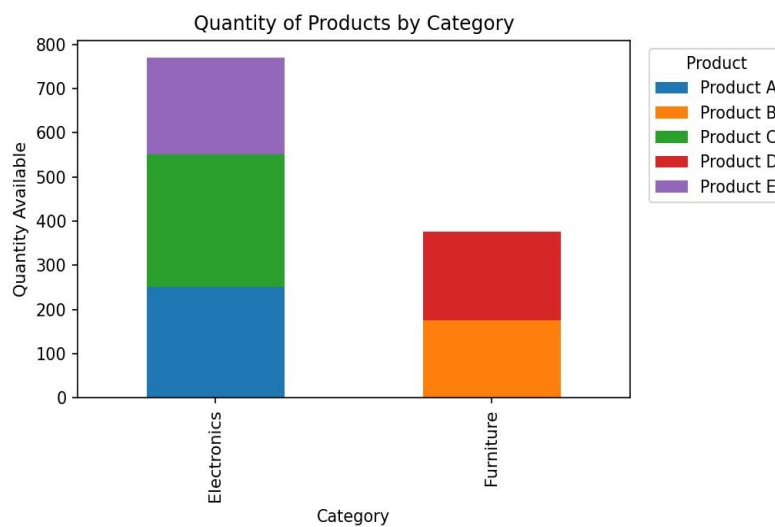
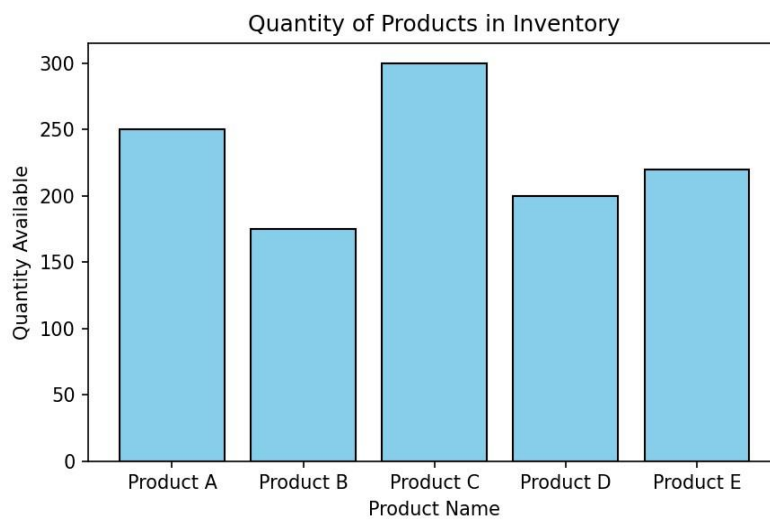
pivot <- xtabs(
  Quantity_Available ~ Category + Product_Name,
  data = inventory
)

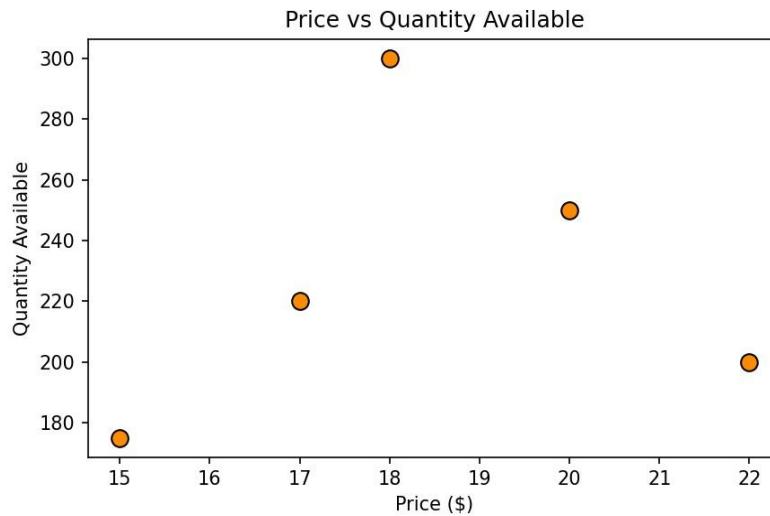
```

```

barplot(
  pivot,
  col = rainbow(nrow(pivot)),
  legend.text = TRUE,
  main = "Quantity of Products by Category",
  xlab = "Product Name",
  ylab = "Quantity Available"
)

```





Set 25: Website Traffic Analysis (R Programming)

```
inventory <- data.frame(
  Product_ID = 1:5,
  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity_Available = c(250, 175, 300, 200, 220),
  Category = c("Electronics", "Furniture", "Electronics", "Furniture", "Electronics"),
  Price = c(20, 15, 18, 22, 17)
)
```

```
barplot(
  inventory$Quantity_Available,
  names.arg = inventory$Product_Name,
  col = "skyblue",
  main = "Quantity of Products in Inventory",
  xlab = "Product Name",
  ylab = "Quantity Available"
)
```

```
pivot <- xtabs(
  Quantity_Available ~ Category + Product_Name,
  data = inventory
)
```

```
barplot(
  pivot,
  col = rainbow(nrow(pivot)),
  legend.text = TRUE,
  main = "Quantity of Products by Category",
  xlab = "Product Name",
  ylab = "Quantity Available"
)
```